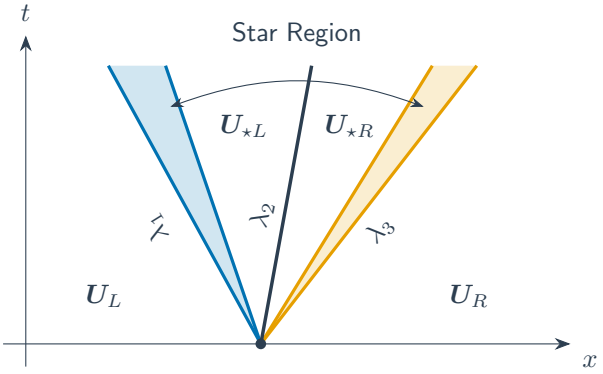
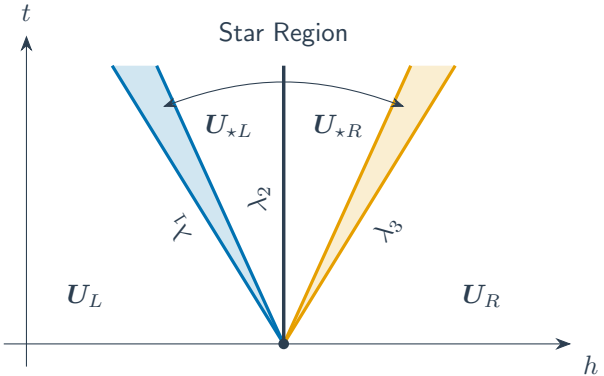


(a) t–x**, Eulerian**

$\lambda_1 = u - c_s \quad \lambda_2 = u \quad \lambda_3 = u + c_s$



(b) t–h**, Lagrangian**

$\lambda_1 = -Z \qquad \lambda_2 = 0 \qquad \lambda_3 = +Z$

All slopes are inverse wave speeds, i.e., $1/\lambda_i$
In panel (b), $\lambda_2 = 0$ sends that slope to infinity.



wedge *open*
rarefaction fan

wedge *closed*
shock

Three waves, always. The wedge width—not the number of lines—indicates the wave type, so this one picture covers all four shock/rarefaction combinations.